TECHNICAL DILIGENCE REVIEW - RECORD 001

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-001

1. Instruction, scope and the evidence relied on

Pinehurst Systems Assurance delivered the present report on 2025-08-19; the scope agreed with Ardent Peak was the service architecture, ownership boundaries and dependency structure of the fleet platform. Report TDR-2025-08 was prepared by Pinehurst Systems Assurance for Ardent Peak and covers the service architecture, ownership boundaries and dependency structure of the fleet platform at Helioscope Robotics, Inc. as at 2025-08-19. For this report the reviewer sampled 3 sites, including the largest deployment and the newest. S. Alderton chose the sample for report TDR-2025-08 after reading the dependency map, so the selection is not proportionate to headcount or to code volume. The reader of the present report should treat each conclusion as bounded by the section it appears in.

Evidence gathering for this report was iterative: an initial document request, 17 interviews, then a second request that closed most of the gaps. In this report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Stated without qualification for once, the conclusion of TDR-2025-08 is that the safety work is the strongest part of the engineering record and the delivery process is the weakest. S. Alderton's conclusion in this report is that the safety work is the strongest part of the engineering record and the delivery process is the weakest. S. Alderton reports management's own position wherever it differs from the reviewer's, and does not merge the two.

Report TDR-2025-08 covers 16 of 34 services; the untested remainder is listed so a reader can see what was not reached. TDR-2025-08 added two components to its sample after the first week, because the interview evidence pointed at them. The reviewer asked in each case whether a declined item had a compensating control, and TDR-2025-08 reports the answer even where it is unsatisfactory. Two positions in the present report are unresolved between the reviewer and the engineering team, and both are left unresolved on the page. S. Alderton has resisted rounding the findings towards a comfortable conclusion in either direction.

Readers of TDR-2025-08 should note that a moderate-evidence finding may carry a high severity, and that this is not a contradiction. The evidence convention in TDR-2025-08 is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Pinehurst Systems Assurance recommends that the 4 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. Post-close monitoring under this report would track four things: objective attainment, change failure rate, open finding ageing and coverage on the control path. S. Alderton has separated what the company has built from what the company can demonstrate about what it has built.

This report relies on the company's incident tracker for completeness of the incident population, which the reviewer could corroborate but not prove. This report does not include a code audit; the reviewer read code where a specific claim required it and no further. In TDR-2025-08 the reviewer records the basis of each conclusion inline, so a reader can disagree with the inference without re-doing the work. S. Alderton treats a backlog ticket as evidence of awareness and a merged change with a test as evidence of remediation, and this report keeps the two apart. S. Alderton has kept the company's terminology even where a different word would have read better, so that a reader can search the source documents.

Interview evidence in report TDR-2025-08 came from 17 sessions across 7 teams, each minuted and circulated for correction. 14 follow-up items arose from the interview programme for TDR-2025-08; 1 were still open at delivery. As instructed, Pinehurst Systems Assurance confined this report to the service architecture, ownership boundaries and dependency structure of the fleet platform; three adjacent questions raised during the work are listed as follow-ups rather than answered here. The subject of report TDR-2025-08 is the service architecture, ownership boundaries and dependency structure of the fleet platform, examined for Ardent Peak by Pinehurst Systems Assurance over 6 weeks ending 2025-08-19. Where the report at TDR-2025-08 uses the word verified, it means the reviewer performed or witnessed something, and nothing weaker.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-08)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-01-001	Identity and Authorization: policy document	policy document	2025-12-03	Meridian Assurance	established provenance
EV-01-002	Fleet Allocator: design record	design record	2026-05-28	Helioscope field reliability	established provenance
EV-01-003	integration tests, cloud: architecture note	architecture note	2025-08-24	the sensor vendor	verified a stated objective
EV-01-004	Wheel Odometry Filter: test report	test report	2026-02-19	the safety controller vendor	verified a stated objective
EV-01-005	Safety Controller: test report	test report	2025-07-30	Helioscope security	verified a stated objective
EV-01-006	Order Broker: measurement extract	measurement extract	2025-09-06	Redwater Security	supported a coverage figure
EV-01-007	Event Bus: test report	test report	2025-06-17	Meridian Assurance	evidenced a remediation
EV-01-008	Fleet Health Monitor: audit output	audit output	2026-03-12	Helioscope engineering	evidenced an approval
EV-01-009	build, robot firmware images: design record	design record	2026-01-27	Meridian Assurance	corroborated an incident narrative
EV-01-010	Telemetry Ingest: measurement extract	measurement extract	2026-07-17	Helioscope security	supported a capacity conclusion
EV-01-011	Billing Usage Meter: design record	design record	2026-03-14	Helioscope security	established provenance
EV-01-012	secret scan: measurement extract	measurement extract	2026-06-29	Helioscope safety	established a dependency
EV-01-013	dependency and licence scan: test report	test report	2025-12-16	the sensor vendor	verified a stated objective
EV-01-014	Simulation Harness: runbook	runbook	2025-11-08	the sensor vendor	established a dependency
EV-01-015	Release Orchestrator: architecture note	architecture note	2025-10-10	the safety controller vendor	corroborated an incident narrative
EV-01-016	Lidar Driver: runbook	runbook	2026-08-19	the safety controller vendor	established a dependency
EV-01-017	Safety Controller: policy document	policy document	2026-02-23	Meridian Assurance	established a dependency
EV-01-018	Simulation Harness: architecture note	architecture note	2025-12-26	Helioscope field reliability	evidenced a remediation
EV-01-019	E-Stop Chain: policy document	policy document	2026-09-23	the safety controller vendor	verified a stated objective
EV-01-020	Motion Planner: release record	release record	2026-02-22	the safety controller vendor	established a dependency
EV-01-021	Motion Planner: incident post-mortem	incident post-mortem	2026-08-03	Redwater Security	established provenance
EV-01-022	Fleet Allocator: vendor statement	vendor statement	2026-03-27	the sensor vendor	evidenced an approval
EV-01-023	Edge Gateway Agent: runbook	runbook	2026-09-14	Redwater Security	evidenced an approval
EV-01-024	build, robot firmware images: test report	test report	2025-10-29	Helioscope engineering	supported a coverage figure
EV-01-025	Release Orchestrator: architecture note	architecture note	2025-12-04	Meridian Assurance	supported a capacity conclusion
EV-01-026	Simulation Harness: configuration export	configuration export	2025-07-12	Helioscope field reliability	supported a capacity conclusion
EV-01-027	Video Snippet Service: interview minute	interview minute	2026-06-29	Helioscope security	evidenced a remediation
EV-01-028	Billing Usage Meter: configuration export	configuration export	2026-06-06	Helioscope security	supported a coverage figure
EV-01-029	Lidar Driver: incident post-mortem	incident post-mortem	2025-11-22	the safety controller vendor	established a dependency
EV-01-030	Event Bus: measurement extract	measurement extract	2025-10-06	Meridian Assurance	evidenced an approval
EV-01-031	Speed and Separation Monitor: architecture note	architecture note	2026-03-15	Helioscope security	evidenced a remediation
EV-01-032	integration tests,	vendor statement	2026-06-28	Helioscope	verified a stated

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	cloud: vendor statement			engineering	objective
EV-01-033	static analysis, robot software: architecture note	architecture note	2025-07-22	Redwater Security	corroborated an incident narrative
EV-01-034	OTA Updater: test report	test report	2026-10-21	Helioscope security	established provenance
EV-01-035	Simulation Harness: runbook	runbook	2026-03-25	Helioscope field reliability	evidenced an approval
EV-01-036	Gripper Controller: interview minute	interview minute	2026-08-27	Helioscope engineering	supported a capacity conclusion

1. Instruction, scope and the evidence relied on - part 2 of 2

The 1. Instruction, scope and the evidence relied on begun above continues from this point, and its row references run consecutively across the parts rather than restarting. Nothing is restated from the earlier part.

Treat the conventions declared for the 1. Instruction, scope and the evidence relied on as governing here too; this part introduces none of its own. No row is omitted at the join.

Schedule E - Index of evidence relied on, with the use made of each item (TDR-2025-08) (continued, part 2 of 2)

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
EV-01-037	Wheel Odometry Filter: test report	test report	2026-05-02	Redwater Security	supported a capacity conclusion
EV-01-038	Identity and Authorization: measurement extract	measurement extract	2026-10-29	Redwater Security	evidenced an approval
EV-01-039	Billing Usage Meter: measurement extract	measurement extract	2026-06-02	Meridian Assurance	established provenance
EV-01-040	simulation suite, short: incident post-mortem	incident post-mortem	2025-05-05	the safety controller vendor	supported a capacity conclusion
EV-01-041	Order Broker: runbook	runbook	2025-09-28	the safety controller vendor	evidenced an approval
EV-01-042	Config Distribution: architecture note	architecture note	2026-07-23	the safety controller vendor	supported a capacity conclusion
EV-01-043	dependency and licence scan: design record	design record	2026-03-03	Meridian Assurance	verified a stated objective
EV-01-044	Event Bus: architecture note	architecture note	2025-08-07	the sensor vendor	supported a capacity conclusion
EV-01-045	Speed and Separation Monitor: test report	test report	2026-05-16	the safety controller vendor	established provenance
EV-01-046	container image vulnerability scan: incident post-mortem	incident post-mortem	2026-01-18	Helioscope field reliability	evidenced a remediation
EV-01-047	Charge Scheduler: architecture note	architecture note	2026-03-12	Helioscope engineering	supported a coverage figure
EV-01-048	Edge Gateway Agent: release record	release record	2026-11-11	the sensor vendor	established a dependency
EV-01-049	WMS Connector: release record	release record	2025-10-01	Helioscope engineering	corroborated an incident narrative
EV-01-050	Config Distribution: release record	release record	2026-10-21	the sensor vendor	verified a stated objective
EV-01-051	performance smoke, cloud: runbook	runbook	2025-08-19	Meridian Assurance	evidenced an approval
EV-01-052	Event Bus: architecture note	architecture note	2026-09-03	the sensor vendor	supported a capacity conclusion
EV-01-053	Motion Planner: architecture note	architecture note	2025-07-21	the safety controller vendor	corroborated an incident narrative
EV-01-054	Fleet Allocator: architecture note	architecture note	2025-07-07	Meridian Assurance	corroborated an incident narrative
EV-01-055	build, cloud container images: incident post-mortem	incident post-mortem	2025-07-25	Helioscope field reliability	evidenced a remediation
EV-01-056	Battery Management Service: release	release record	2026-06-20	Redwater Security	supported a coverage figure

Evidence ref	Document or artefact	Type	Date	Provided by	Relied on for
	record				
EV-01-057	simulation suite, nightly: measurement extract	measurement extract	2026-07-24	Helioscope field reliability	evidenced an approval
EV-01-058	performance smoke, cloud: audit output	audit output	2025-06-02	the sensor vendor	corroborated an incident narrative
EV-01-059	static analysis, robot software: policy document	policy document	2026-04-20	Helioscope field reliability	evidenced an approval
EV-01-060	static analysis, robot software: release record	release record	2026-05-28	the sensor vendor	verified a stated objective
EV-01-061	performance smoke, cloud: interview minute	interview minute	2026-11-03	Redwater Security	established provenance
EV-01-062	Battery Management Service: architecture note	architecture note	2025-08-12	the sensor vendor	established provenance
EV-01-063	Order Broker: architecture note	architecture note	2025-12-24	Helioscope security	evidenced an approval
EV-01-064	lint and format: audit output	audit output	2025-09-21	the safety controller vendor	evidenced a remediation
EV-01-065	Wheel Odometry Filter: vendor statement	vendor statement	2026-06-25	Helioscope engineering	verified a stated objective
EV-01-066	Costmap Fusion: policy document	policy document	2026-09-11	Helioscope security	established a dependency
EV-01-067	Localization Engine: architecture note	architecture note	2026-05-18	Redwater Security	established provenance
EV-01-068	secret scan: architecture note	architecture note	2025-06-14	Helioscope security	established provenance
EV-01-069	Site Provisioning Service: configuration export	configuration export	2026-04-04	Helioscope safety	established a dependency
EV-01-070	Charge Scheduler: audit output	audit output	2026-03-14	Meridian Assurance	verified a stated objective
EV-01-071	Wheel Odometry Filter: interview minute	interview minute	2025-10-21	Helioscope engineering	established a dependency
EV-01-072	Costmap Fusion: runbook	runbook	2026-07-18	Helioscope engineering	verified a stated objective

2. Simulation Harness and the services on the control path

Interface discipline around Simulation Harness is adequate; the schema is versioned and change is gated on a compatibility check. Simulation Harness pairs a Python 3.11 with a C++ physics core implementation with developer workstations and a batch cluster, and its data lives in object storage for scenario definitions and run artefacts. Evidence examined for Simulation Harness comprised the scenario catalogue, two weeks of nightly run records and a determinism experiment. Upstream of Simulation Harness sit the planner and localization builds under test; downstream, the release orchestrator gate and the autonomy regression report consume its output. Simulation Harness has a single named owner, Colin Achebe of Developer Productivity, which is not true of every service reviewed. Simulation Harness is included because a reader of TDR-2025-08 should be able to disagree with the inference without repeating the work.

Simulation Harness ships continuous, from the main branch; the reviewer sampled the last twelve releases. 31 defects remain open against Simulation Harness; the one the reviewer pressed on is that scenario determinism breaks when the physics core is rebuilt with a different optimisation level. On end-of-life exposure, Simulation Harness shows the physics core depends on a library branch that stopped receiving fixes in 2023. Coverage for Simulation Harness was 47 per cent line coverage at the review date, and the trend over the reviewed period was up by seven points. Simulation Harness commits that the nightly scenario suite completes inside four hours, and the objective is measured continuously rather than sampled. Simulation Harness carries a follow-up request that was satisfied after the draft of TDR-2025-08 was circulated.

Data ownership is clear for Payload Scale Interface: it writes a local calibration record and no other service writes there. Payload Scale Interface shows 36 per cent line coverage in the pipeline, and the reviewer regards that as adequate for the risk this component carries. Supportability of Payload Scale Interface turns on one fact: none identified. Payload Scale Interface sits between the load cell hardware and the gripper controller and the order execution path, and that position is what makes its availability objective load-bearing. Placement of Payload Scale Interface is edge, only on robots fitted with the weighing deck, which the reviewer tested by tracing a request end to end. Payload Scale Interface has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

Payload Scale Interface releases as needed. Payload Scale Interface has a single named owner, Ines Duarte of Robot Firmware, which is not true of every service reviewed. Evidence examined for Payload Scale Interface comprised the interface specification, calibration records for nine units and a module-swap test. A local calibration record is the system of record for Payload Scale Interface, a C++17 component running on on-robot Linux, user-space driver. Of the 11 open items against Payload Scale Interface, the reviewer treats one as material: the calibration record is stored locally and is lost when the compute module is swapped. Payload Scale Interface is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Fleet Allocator has a single named owner, Marta Kellerman of Fleet Services, which is not true of every service reviewed. Behind Fleet Allocator sits PostgreSQL 14 with a synchronous standby; the service itself is Go 1.22 on cloud, containers on a managed Kubernetes cluster. For Fleet Allocator the supportability position is none identified in the reviewed manifests. Fleet Allocator was checked for shared-database coupling, and the ownership boundary held. Test coverage on Fleet Allocator is 82 per cent line coverage and the measurement basis was confirmed with Marta Kellerman. Fleet Allocator is the clearest instance in TDR-2025-08 of a control that exists on paper and is weaker in practice.

Evidence examined for Fleet Allocator comprised the allocator service contract, two design reviews and a week of assignment audit logs. Fleet Services publishes one objective for Fleet Allocator: task assignment inside 900 ms of order arrival at p95. The order broker and the robot shadow store feed Fleet Allocator; a failure in Fleet Allocator is felt by the traffic manager, the charge scheduler and the fleet health monitor. 9 defects remain open against Fleet Allocator; the one the reviewer pressed on is that assignment starvation for robots parked at chargers below the battery reserve threshold. The reviewer's judgement on Fleet Allocator is that the reviewer accepts the engineering position while noting that the verifying test does not exist. Fleet Allocator has no single accountable owner, and that is the finding the reviewer would press hardest.

Battery Management Service publishes a stable interface, though two fields used by the charge scheduler and the fleet health monitor are documented only in code comments. Robot Firmware publishes one objective for Battery Management Service: pack cell imbalance is reported before it reaches the derating threshold. Battery Management Service releases pack firmware twice a year, the cloud service monthly. 10 items sit in the Battery Management Service backlog; the entry that bears on the investment case is that state-of-health estimates drift on packs that spend most of their life on opportunity charge. Battery Management Service sits at 54 per cent line coverage; the trend across the reviewed period was up by three points. Battery Management Service has no single accountable owner, and that is the finding the reviewer would press hardest.

The dependency map places the pack sensors and the charge scheduler above Battery Management Service and the charge scheduler and the fleet health monitor below it. For Battery Management Service the reviewer records that the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. Behind Battery Management Service sits pack non-volatile memory and a cloud time-series store; the service itself is C on the pack controller, Python 3.11 in the cloud on split: pack firmware and a cloud analytics service. The manifests place Battery Management Service split edge and cloud. Battery Management Service is owned by the Robot Firmware team, with Peter Halvorsen named as the engineering owner of record. Battery Management Service is reported here in full because a summary of it in an earlier draft of TDR-2025-08 was misread as an assurance.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-08)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th	the costmap fusion pipeline and the	none; the toolchain is pinned to a

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				percentile	localization engine	supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within 250 ms of button actuation on every tested unit	the operator stop buttons and the perimeter interlocks	none; the relay parts are current production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in mapped aisles at survey speed	the lidar driver, the wheel odometry filter and the map distribution channel	an unmaintained scan-matching library is still linked in the fallback path
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and Authorization	Go 1.22	Security Engineering	84%	token issuance inside 200 ms at p95 and no unauthenticated write path	the customer directory federations and the secret store	none identified in the reviewed manifests

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

2. Simulation Harness and the services on the control path - part 2 of 2

Below, the 2. Simulation Harness and the services on the control path continues; nothing separates this part from the last except the page on which it happens to fall.

Column definitions stated with the first part of the 2. Simulation Harness and the services on the control path govern the whole of it, and any qualification recorded there is carried forward here. Pagination alone determines where it falls.

Schedule A - Service inventory, ownership and objective attainment (TDR-2025-08) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow Store	Go 1.22	Fleet Services	68%	shadow state is no more than two seconds behind	the telemetry ingest path and the edge gateway	none identified

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				the robot at p95	agent	
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four hours	the on-robot services that emit telemetry	none; the transport library was updated in early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production without a recorded approval and a passing gate set	the build system, the simulation harness gate and the identity service	none identified

3. Deployment topology and failure domains, beginning with Config Distribution

Config Distribution is deployed split edge and cloud, with a signed bundle format. On end-of-life exposure, Config Distribution shows none; the bundle format was revised in 2025. Config Distribution was assessed against the objective and the defect record, and the reviewer accepts the engineering position while noting that the verifying test does not exist. Config Distribution was checked for shared-database coupling, and the ownership boundary held. Content-addressed object storage with a PostgreSQL 14 index is the system of record for Config Distribution, a Rust 1.79 agent with a Go 1.22 control service component running on split: an on-robot agent and a cloud service. Config Distribution was tested rather than accepted, and the test is described in the working papers behind TDR-2025-08.

Where Config Distribution is concerned, the commitment is that a configuration change reaches every online robot inside five minutes; the reviewer traced the metric to its source. Documents and data inspected for Config Distribution were the bundle format specification, distribution latency measurements and an offline apply test. The open defect backlog for Config Distribution holds 6 items, the most consequential being that a robot that is offline at publication applies the change without re-validating preconditions. Reading the dependency map from Config Distribution outward

gives the site provisioning service and the artefact signing service on the input side and the on-robot supervisor and the motion planner parameters on the output side. Config Distribution reports 72 per cent line coverage, measured by the same instrumentation the pipeline uses. Config Distribution is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The reviewer's judgement on Site Provisioning Service is that the component would not block the transaction, and it would appear on any competent post-close engineering plan. Hosting for Site Provisioning Service is cloud, containers on a managed Kubernetes cluster, and its Go 1.22 code writes through to PostgreSQL 14 with a configuration history table. Upstream of Site Provisioning Service sit the identity service and the customer network intake records; downstream, the config distribution service and the WMS connector consume its output. Site Provisioning Service reports 64 per cent line coverage, measured by the same instrumentation the pipeline uses. Platform Infrastructure publishes one objective for Site Provisioning Service: a new site reaches a working configuration inside one working day. Site Provisioning Service is reported here in full because a summary of it in an earlier draft of TDR-2025-08 was misread as an assurance.

One coupling concern applies to Site Provisioning Service: PostgreSQL 14 with a configuration history table is shared with a second service, so a schema change is not a local decision. On paper Site Provisioning Service releases fortnightly, and the observed practice did not diverge from that. Platform Infrastructure tracks 5 open defects on Site Provisioning Service, including the acknowledged case where site templates diverged after two customers required bespoke network settings. Site Provisioning Service's dependency ageing was checked: none identified. Site Provisioning Service is deployed cloud, global. Site Provisioning Service was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Safety Engineering holds Safety Controller, and the ownership entry naming Bettina Vogel matches the on-call rotation. Safety Controller depends on the E-stop chain, the speed and separation monitor and the drive current sensors and is depended on by the drive power contactors and the robot fault annunciator. Safety Controller ships twice a year, with full revalidation of the safety case; the reviewer sampled the last twelve releases. The interface Safety Controller publishes is versioned, and the drive power contactors and the robot fault annunciator were checked for use of any undocumented field. For Safety Controller the stated target is that a validated protective-stop trigger de-energises the drive inside 120 ms, expressed against production traffic rather than a benchmark. Safety Controller is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Behind Safety Controller sits a non-volatile fault log on the controller; the service itself is C, written to a constrained coding subset on a safety-rated controller module supplied by a third party. Safety Controller was examined for unsupported components; none; the supplier's support commitment runs to 2031. Evidence examined for Safety Controller comprised the supplier assessment report, the internal safety case and eight bench trigger tests. Taking the evidence together, Safety Controller it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Safety Controller reports 91 per cent line coverage, measured by the same instrumentation the pipeline uses. Safety Controller rests on a single source, and the reviewer has said so rather than implying corroboration.

Against Analytics Warehouse Loader, the coverage figure is 45 per cent line coverage, which coverage alone overstates the assurance here, because the bench suite is the real safety net. Python 3.11 with SQL transformations is the implementation language for Analytics Warehouse Loader, which is hosted on cloud, a scheduled batch runner with a columnar warehouse with daily partitions behind it. Analytics Warehouse Loader is owned by the Data and Telemetry team, with Farida Cisse named as the engineering owner of record. Data and Telemetry publishes one objective for Analytics Warehouse Loader: the daily load completes before 06:00 local time at every region. Data and Telemetry tracks 20 open defects on Analytics Warehouse Loader, including the acknowledged case where a late-arriving telemetry partition is silently excluded rather than reprocessed. Analytics Warehouse Loader was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

Reading the dependency map from Analytics Warehouse Loader outward gives the telemetry ingest path, the order broker and the billing usage meter on the input side and the customer reporting views and the internal metrics layer on the output side. Analytics Warehouse Loader rests on the load schedule, partition completeness checks and two months of load audit records so far as this review is concerned. On paper Analytics Warehouse Loader releases weekly, and the

observed practice did not diverge from that. The interface Analytics Warehouse Loader publishes is versioned, and the customer reporting views and the internal metrics layer were checked for use of any undocumented field. Read against its objective, Analytics Warehouse Loader the reviewer accepts the engineering position while noting that the verifying test does not exist. Analytics Warehouse Loader was one of two items where TDR-2025-08 could not obtain the underlying data within the engagement.

Release cadence for Robot Shadow Store is weekly, and the release records bear that out. In production Robot Shadow Store is cloud, regional, sharded by site, unchanged since the last architecture review. Against Robot Shadow Store, the coverage figure is 68 per cent line coverage, which the reviewer notes that the untested share includes the error handling. The fleet allocator, the traffic manager and the remote operations console rely on Robot Shadow Store, which in turn relies on the telemetry ingest path and the edge gateway agent. Read against its objective, Robot Shadow Store it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. Robot Shadow Store was inspected in the repository rather than described from the ticket, which is why the account is specific.

Go 1.22 is the implementation language for Robot Shadow Store, which is hosted on cloud, containers with a sharded cache with Redis 7 as the hot path with PostgreSQL 14 as the durable record behind it. Robot Shadow Store was tested against the dependency ageing register, which records none identified. The Fleet Services team carries Robot Shadow Store on its service inventory, and Marta Kellerman answered for it in interview. Service-level intent for Robot Shadow Store states that shadow state is no more than two seconds behind the robot at p95. Of the 9 open items against Robot Shadow Store, the reviewer treats one as material: a shard rebalance briefly serves stale shadow state without marking it stale. Robot Shadow Store is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Schedule B - Deployment topology and failure domain by service (TDR-2025-08)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Motion Planner	edge, one instance per robot with no cloud fallback	every three weeks, promoted ring by ring	1	one robot	Priya Raghunathan
Costmap Fusion	edge, co-scheduled with the planner on the compute module	monthly, batched with perception model updates	4	one robot	Daniel Okafor
Fleet Allocator	cloud, one regional deployment serving all sites in that region	weekly, behind a change approval	6	one region	Marta Kellerman
Traffic Manager	cloud, regional, with a degraded on-robot yield behaviour if the link drops	weekly, paired with the allocator release	5	one region	Owen Brantley
Charge Scheduler	cloud, regional, with a conservative local fallback policy on the robot	fortnightly, without a formal approval gate until March 2026	1	one region	Nadia Semenova
Safety Controller	edge, dual-channel, with cross-checking between channels	twice a year, with full revalidation of the safety case	4	one robot	Bettina Vogel
E-Stop Chain	edge, hardwired, independent of the compute module	changed only by an engineering change order	2	one robot	Theo Bassett
Lidar Driver	edge, one process per sensor	as needed, historically twice a year	5	one robot	Yuki Harada
Localization Engine	edge, with map distribution from the cloud	every three weeks with the planner	4	one robot	Rahul Menon
WMS Connector	cloud, one deployment per customer tenant	every two weeks, per tenant on request	6	one region	Tomas Villareal
Order Broker	cloud, regional	weekly	6	one region	Grace Lindqvist
Telemetry Ingest	cloud, one ingest endpoint per region with an edge buffer on the robot	weekly, with schema changes gated on a compatibility check	3	one robot	Farida Cisse
OTA Updater	split edge and cloud,	every three weeks,	1	one robot	Erik Palmgren

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
	with staged rings and a per-site pause control	following the autonomy release			
Identity and Authorization	cloud, global control plane with regional caches	weekly, with a security review on every change	2	one region	Sonia Krall
Remote Operations Console	cloud, with a thin session relay in each region	twice a week	1	one region	Anjali Sekhon
Simulation Harness	cloud batch, with a local mode for developers	continuous, from the main branch	3	one region	Colin Achebe
Hardware-in-the-Loop Rig	laboratory, two rigs, one of which lags the current sensor configuration	the harness changes with each firmware release	4	one region	Nathan Oduya

3. Deployment topology and failure domains, beginning with Config Distribution - part 2 of 2

This is part 2 of the 3. Deployment topology and failure domains, beginning with Config Distribution, carried over unbroken; a reader reconciling a reference should treat the parts as a single register. Nothing is restated from the earlier part.

Readers should carry the 3. Deployment topology and failure domains, beginning with Config Distribution's stated conventions across the break, since this part adds no definition of its own. No row is omitted at the join.

Schedule B - Deployment topology and failure domain by service (TDR-2025-08) (continued, part 2 of 2)

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Fleet Health Monitor	cloud, regional	fortnightly	1	one region	Wanda Njoku
Battery Management Service	split edge and cloud	pack firmware twice a year, the cloud service monthly	1	one robot	Peter Halvorsen
Gripper Controller	edge, only on the two robot variants fitted with a manipulator	quarterly	3	one robot	Ines Duarte
Map Authoring Studio	cloud, single global deployment	weekly	3	one region	Lucia Ferrante
Site Provisioning Service	cloud, global	fortnightly	1	one region	Marcus Duplessis
Event Bus	cloud, one cluster per region, three availability zones	upgraded twice a year	6	one region	Erik Palmgren
Config Distribution	split edge and cloud, with a signed bundle format	weekly	4	one robot	Sam Oyelaran
Robot Shadow Store	cloud, regional, sharded by site	weekly	1	one region	Marta Kellerman
Analytics Warehouse Loader	cloud, one loader per region writing to a shared warehouse	weekly	5	one region	Farida Cisse
Billing Usage Meter	cloud, regional	monthly, aligned to the billing cycle	4	one region	Nadia Semenova
Alarm Router	cloud, regional, with a second delivery provider	fortnightly	4	one region	Wanda Njoku
Video Snippet Service	cloud, regional, with an on-robot capture buffer	monthly	2	one region	Anjali Sekhon
Speed and Separation Monitor	edge, cross-checked against the safety controller's own inputs	twice a year, with revalidation	2	one robot	Bettina Vogel
Wheel Odometry Filter	edge	with the firmware, twice a year	2	one robot	Yuki Harada
Payload Scale Interface	edge, only on robots fitted with the weighing deck	as needed	1	one robot	Ines Duarte

Service	Deployment topology	Release cadence	Instances per region	Failure domain	Engineering owner
Edge Gateway Agent	edge, one per robot, with mutual authentication to the cloud	every three weeks	4	one robot	Sam Oyelaran
Release Orchestrator	cloud, single global deployment	continuous, self-hosting	5	one region	Colin Achebe

4. Interface boundaries and data ownership around Localization Engine

Against Localization Engine the backlog runs to 16 items, and it includes a known behaviour whereby relocalisation after a lift-and-place takes longer than the operator prompt allows. The reviewer's judgement on Localization Engine is that the risk is concentrated in one code path that has neither automated coverage nor a bench equivalent. Inputs to Localization Engine come from the lidar driver, the wheel odometry filter and the map distribution channel, and its consumers are the motion planner and the fleet health monitor. 66 per cent line coverage is reported for Localization Engine, and the reviewer re-ran the measurement on a clean checkout. The objective recorded against Localization Engine is that pose error stays under 25 mm in mapped aisles at survey speed, and it is reported weekly to the engineering leadership. Localization Engine is the reason one section of TDR-2025-08 concludes narrowly rather than generally.

On paper Localization Engine releases every three weeks with the planner, and the observed practice did not diverge from that. Localization Engine's dependency ageing was checked: an unmaintained scan-matching library is still linked in the fallback path. The topology for Localization Engine is edge, with map distribution from the cloud, and the reviewer confirmed it against the deployment manifests. A local map cache with a periodic checkpoint is the system of record for Localization Engine, a C++17 with a Rust bridge layer component running on on-robot Linux, pinned to a dedicated core. Localization Engine was checked for shared-database coupling, and the ownership boundary held. Localization Engine is treated as material to the investment case rather than as a routine engineering item.

Evidence examined for Map Authoring Studio comprised the authoring workflow document, the revision history for six sites and an export audit. Hosting for Map Authoring Studio is cloud, a desktop-class web application, and its TypeScript with a Python service layer code writes through to PostgreSQL 14 with map revisions in object storage. On end-of-life exposure, Map Authoring Studio shows a geometry rendering library one major version behind its supported line. Map Authoring Studio depends on the site survey imports and the identity service and is depended on by the map distribution channel and the localization engine. Ownership of Map Authoring Studio sits with Applications; Lucia Ferrante signs off changes to it. Map Authoring Studio is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Map Authoring Studio carries 22 open defects, among them a case where an unpublished draft can be exported and hand-loaded, bypassing the review step. The objective recorded against Map Authoring Studio is that a published map revision reaches every robot at the site inside fifteen minutes, and it is reported weekly to the engineering leadership. The topology for Map Authoring Studio is cloud, single global deployment, and the reviewer confirmed it against the deployment manifests. Map Authoring Studio releases weekly. Map Authoring Studio was checked for shared-database coupling, and the ownership boundary held. Map Authoring Studio is reported at the evidence strength available and has not been promoted to fit the narrative.

Against Wheel Odometry Filter, the coverage figure is 62 per cent line coverage, which the reviewer notes that the untested share includes the error handling. The published objective for Wheel Odometry Filter is that odometry drift stays under 1.5 per cent of distance travelled on a clean floor. Wheel Odometry Filter is written in C++17, executes on on-robot Linux, in the drive control process, and keeps its state in no persistent store. Wheel Odometry Filter owns its own data and exposes it only through a versioned interface, which the reviewer read. Wheel Odometry Filter carries 8 open defects, among them a case where drift exceeds the specification on wet or dusty floors and is not flagged to the operator. Wheel Odometry Filter is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Wheel Odometry Filter was assessed on the filter design note, drift measurements at four sites and a wet-floor bench run, and nothing beyond that was available in the data room. On end-of-life exposure, Wheel Odometry Filter shows none identified. Read against its objective, Wheel Odometry Filter it is fit for the current fleet size, and the

reviewer found no evidence that it would remain so at twice that scale. The wheel encoders and the inertial measurement unit feed Wheel Odometry Filter; a failure in Wheel Odometry Filter is felt by the localization engine and the motion planner. Yuki Harada of Robot Firmware maintains Wheel Odometry Filter, and the service inventory entry was current at the review date. Wheel Odometry Filter is included because a reader of TDR-2025-08 should be able to disagree with the inference without repeating the work.

The Safety Engineering team carries E-Stop Chain on its service inventory, and Theo Bassett answered for it in interview. E-Stop Chain runs edge, hardwired, independent of the compute module; that placement is what determines whether a fault is one robot or one region. For E-Stop Chain the stated target is that drive power is removed within 250 ms of button actuation on every tested unit, expressed against production traffic rather than a benchmark. On paper E-Stop Chain releases changed only by an engineering change order, and the observed practice did not diverge from that. E-Stop Chain sits between the operator stop buttons and the perimeter interlocks and the safety controller and the drive contactors, and that position is what makes its availability objective load-bearing. E-Stop Chain would be a candidate for the first post-close engineering review under any reasonable plan.

Hosting for E-Stop Chain is a safety-rated hardware assembly, no general-purpose operating system, and its VHDL with discrete relay logic code writes through to no persistent store; events are latched to the safety controller. 2 defects remain open against E-Stop Chain; the one the reviewer pressed on is that one pre-production harness revision remains in the field on nine robots at two sites. E-Stop Chain sits at no meaningful automated unit coverage, verification being by witnessed test instead; the trend across the reviewed period was up by seven points. Taking the evidence together, E-Stop Chain the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. For E-Stop Chain the reviewer worked from the wiring diagrams, the change order file and a witnessed actuation test on four units. E-Stop Chain rests on a single source, and the reviewer has said so rather than implying corroboration.

On end-of-life exposure, Hardware-in-the-Loop Rig shows the rig controller runs an operating system release that left vendor support in 2025. Nathan Oduya of Robot Firmware maintains Hardware-in-the-Loop Rig, and the service inventory entry was current at the review date. Hardware-in-the-Loop Rig was assessed against the objective and the defect record, and it is fit for the current fleet size, and the reviewer found no evidence that it would remain so at twice that scale. The evidence base for Hardware-in-the-Loop Rig was the bench suite definition, rig configuration inventory and two firmware release records. Hardware-in-the-Loop Rig shows 39 per cent line coverage in the pipeline, and the number is credible and the measurement basis is honest. Hardware-in-the-Loop Rig was reviewed against the position twelve months earlier, and nothing measurable has changed.

Data ownership is clear for Hardware-in-the-Loop Rig: it writes a local results database, backed up weekly and no other service writes there. Robot Firmware releases Hardware-in-the-Loop Rig the harness changes with each firmware release, and no release in the sampled window bypassed the gate set. The open defect backlog for Hardware-in-the-Loop Rig holds 13 items, the most consequential being that the second rig carries a superseded lidar variant, so one sensor path is only tested on one rig. The firmware release decision record rely on Hardware-in-the-Loop Rig, which in turn relies on the firmware candidate artefacts and the bench sensor stimulators. Hardware-in-the-Loop Rig was built in Python 3.11 test harness around production firmware and has not been rewritten; it still runs on bench hardware in the Fremont laboratory against a local results database, backed up weekly. Hardware-in-the-Loop Rig was excluded from an earlier draft of TDR-2025-08 and reinstated after the reviewer obtained the supporting record.

Schedule C - Producer and consumer interface contracts examined (TDR-2025-08)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Motion Planner	traffic manager	internal protocol	v2.8	enforced in the pipeline	2 in the reviewed period
Motion Planner	on-robot drive supervisor	request-response interface	v2.7	enforced in the pipeline	2 in the reviewed period
Costmap Fusion	motion planner	published schema	v3.9	not automated	2 in the reviewed period
Costmap Fusion	speed	streamed topic	v2.10	enforced in the pipeline	2 in the reviewed period
Fleet Allocator	traffic manager	request-response interface	v4.10	not automated	2 in the reviewed period
Fleet Allocator	fleet health monitor	shared event contract	v1.7	enforced in the	1 in the reviewed

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
				pipeline	period
Traffic Manager	motion planner	shared event contract	v3.4	advisory only	2 in the reviewed period
Traffic Manager	remote operations console	streamed topic	v3.4	enforced in the pipeline	none
Charge Scheduler	fleet allocator	published schema	v2.6	not automated	1 in the reviewed period
Charge Scheduler	fleet health monitor	request-response interface	v1.8	not automated	1 in the reviewed period
Safety Controller	drive power contactors	published schema	v2.5	advisory only	1 in the reviewed period
Safety Controller	robot fault annunciator	streamed topic	v1.2	enforced on the producer side	none
E-Stop Chain	safety controller	internal protocol	v4.9	not automated	1 in the reviewed period
E-Stop Chain	drive contactors	shared event contract	v4.5	not automated	1 in the reviewed period
Lidar Driver	costmap fusion pipeline	streamed topic	v4.8	advisory only	none
Lidar Driver	localization engine	published schema	v3.1	advisory only	none
Localization Engine	motion planner	streamed topic	v2.0	enforced on the producer side	2 in the reviewed period
Localization Engine	fleet health monitor	published schema	v2.7	not automated	1 in the reviewed period
WMS Connector	order broker	streamed topic	v4.1	not automated	1 in the reviewed period
WMS Connector	billing usage meter	shared event contract	v1.2	not automated	2 in the reviewed period
Order Broker	fleet allocator	request-response interface	v1.4	enforced in the pipeline	2 in the reviewed period
Order Broker	analytics warehouse loader	published schema	v4.11	enforced on the producer side	2 in the reviewed period
Telemetry Ingest	fleet health monitor	internal protocol	v3.8	enforced in the pipeline	1 in the reviewed period
Telemetry Ingest	alarm router	request-response interface	v2.7	enforced in the pipeline	1 in the reviewed period
OTA Updater	on-robot supervisor	shared event contract	v1.7	not automated	none
OTA Updater	fleet health monitor	streamed topic	v2.10	enforced on the producer side	1 in the reviewed period
Identity and Authorization	every cloud service	shared event contract	v4.10	enforced on the producer side	2 in the reviewed period
Identity and Authorization	remote operations console	published schema	v1.8	advisory only	1 in the reviewed period
Remote Operations Console	traffic manager	request-response interface	v2.11	not automated	none
Remote Operations Console	alarm router	published schema	v3.0	enforced on the producer side	2 in the reviewed period
Simulation Harness	release orchestrator gate	request-response interface	v4.10	enforced on the producer side	2 in the reviewed period
Simulation Harness	autonomy regression report	request-response interface	v2.4	not automated	none
Hardware-in-the-Loop Rig	firmware release decision record	published schema	v2.3	enforced on the producer side	1 in the reviewed period

4. Interface boundaries and data ownership around Localization Engine - part 2 of 2

Set out next is the balance of the 4. Interface boundaries and data ownership around Localization Engine, in the sequence in which the rows were originally compiled. Nothing is restated from the earlier part.

Scope, cut-off and units for the 4. Interface boundaries and data ownership around Localization Engine were fixed at its first part and are not repeated, being unaltered. The sequence is unbroken either side of it.

Schedule C - Producer and consumer interface contracts examined (TDR-2025-08) (continued, part 2 of 2)

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Fleet Health Monitor	alarm router	streamed topic	v4.10	enforced on the producer side	none
Fleet Health Monitor	field service workflow	published schema	v4.6	advisory only	2 in the reviewed period
Battery Management	charge scheduler	request-response	v4.11	not automated	none

Producer service	Consumer	Interface form	Interface version	Compatibility check	Breaking changes
Service		interface			
Battery Management Service	fleet health monitor	streamed topic	v2.1	enforced on the producer side	2 in the reviewed period
Gripper Controller	order execution path	published schema	v2.4	enforced in the pipeline	none
Gripper Controller	safety controller fault input	published schema	v3.5	enforced on the producer side	1 in the reviewed period
Map Authoring Studio	map distribution channel	internal protocol	v3.4	advisory only	none
Map Authoring Studio	localization engine	internal protocol	v1.8	not automated	2 in the reviewed period
Site Provisioning Service	config distribution service	published schema	v1.6	enforced on the producer side	none
Site Provisioning Service	WMS connector	internal protocol	v2.10	enforced in the pipeline	1 in the reviewed period
Event Bus	every consuming service	published schema	v1.5	advisory only	none
Config Distribution	on-robot supervisor	streamed topic	v2.1	enforced on the producer side	none
Config Distribution	motion planner parameters	internal protocol	v1.3	enforced on the producer side	none
Robot Shadow Store	fleet allocator	shared event contract	v4.9	enforced on the producer side	none
Robot Shadow Store	remote operations console	streamed topic	v1.9	enforced on the producer side	none
Analytics Warehouse Loader	customer reporting views	streamed topic	v4.2	enforced in the pipeline	none
Analytics Warehouse Loader	internal metrics layer	shared event contract	v2.1	enforced on the producer side	2 in the reviewed period
Billing Usage Meter	finance export	streamed topic	v2.7	enforced in the pipeline	none
Billing Usage Meter	customer usage report	streamed topic	v3.8	advisory only	1 in the reviewed period
Alarm Router	paging provider	streamed topic	v2.3	not automated	none
Alarm Router	customer notification channel	request-response interface	v3.6	not automated	2 in the reviewed period
Video Snippet Service	remote operations console	shared event contract	v2.5	enforced in the pipeline	2 in the reviewed period
Video Snippet Service	incident review workflow	shared event contract	v3.9	enforced in the pipeline	1 in the reviewed period
Speed and Separation Monitor	safety controller	streamed topic	v1.3	enforced in the pipeline	none
Speed and Separation Monitor	motion planner speed limit	published schema	v1.6	advisory only	2 in the reviewed period
Wheel Odometry Filter	localization engine	request-response interface	v1.2	not automated	2 in the reviewed period
Wheel Odometry Filter	motion planner	published schema	v2.7	enforced on the producer side	1 in the reviewed period
Payload Scale Interface	gripper controller	request-response interface	v2.0	enforced on the producer side	2 in the reviewed period
Payload Scale Interface	order execution path	shared event contract	v4.9	not automated	1 in the reviewed period
Edge Gateway Agent	telemetry ingest path	streamed topic	v4.3	enforced in the pipeline	none
Edge Gateway Agent	robot shadow store	streamed topic	v2.1	advisory only	2 in the reviewed period
Release Orchestrator	OTA updater	request-response interface	v1.4	enforced in the pipeline	1 in the reviewed period
Release Orchestrator	cloud deployment controllers	internal protocol	v4.0	advisory only	1 in the reviewed period

5. Automated test coverage across the reviewed services

Motion Planner is promoted every three weeks, promoted ring by ring, with the ring order fixed by policy. Interface discipline around Motion Planner is adequate; the schema is versioned and change is gated on a compatibility check. Service-level intent for Motion Planner states that plan cycle completes inside 40 ms at the 99th percentile. On Motion Planner the conclusion is qualified: the component would not block the transaction, and it would appear on any competent post-close engineering plan. The backlog for Motion Planner is 14 items deep and contains an acknowledged defect

whereby replan oscillation when two robots negotiate the same narrow-aisle pinch point. Motion Planner was excluded from an earlier draft of TDR-2025-08 and reinstated after the reviewer obtained the supporting record.

Implementation of Speed and Separation Monitor rests on C++17 with a safety-reviewed core; the process runs on on-robot Linux, pinned to the safety compute path and persists to a rolling in-memory window with fault snapshots to the safety controller. Supportability of Speed and Separation Monitor turns on one fact: none; the sensing hardware is current production. Of the 4 open items against Speed and Separation Monitor, the reviewer treats one as material: the protective distance calculation uses a fixed operator approach speed rather than a measured one. The Safety Engineering team carries Speed and Separation Monitor on its service inventory, and Bettina Vogel answered for it in interview. Inputs to Speed and Separation Monitor come from the costmap fusion pipeline and the safety-rated scanner inputs, and its consumers are the safety controller and the motion planner speed limit. Speed and Separation Monitor was cross-checked against the engineering record supplied for TDR-2025-08, and the two agree.

Platform Infrastructure holds OTA Updater, and the ownership entry naming Erik Palmgren matches the on-call rotation. For OTA Updater the supportability position is none; artefact signing was moved to the current key format in 2025. What the reviewer had for OTA Updater was the update state machine specification, signing key custody notes and two rollback drills; the conclusions below do not reach past it. Platform Infrastructure tracks 8 open defects on OTA Updater, including the acknowledged case where a paused ring resumes automatically after a service restart, which surprised two site teams. Reading the dependency map from OTA Updater outward gives the release orchestrator and the artefact signing service on the input side and the on-robot supervisor and the fleet health monitor on the output side. OTA Updater is treated as material to the investment case rather than as a routine engineering item.

One coupling concern applies to Event Bus: the broker's own log storage with seven-day retention is shared with a second service, so a schema change is not a local decision. Platform Infrastructure tracks 4 open defects on Event Bus, including the acknowledged case where one topic carries both control and telemetry traffic, so a telemetry backlog delays control events. Event Bus is deployed cloud, one cluster per region, three availability zones. The objective recorded against Event Bus is that publish acknowledgement inside 25 ms at p95 and no partition without a leader, and it is reported weekly to the engineering leadership. Test coverage on Event Bus is no meaningful automated unit coverage, verification being by witnessed test instead and the measurement basis was confirmed with Erik Palmgren. Event Bus was reviewed against the position twelve months earlier, and nothing measurable has changed.

Schedule D - Automated coverage trend on the reviewed services (TDR-2025-08)

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Motion Planner	63%	70%	71%	+8	Autonomy Core
Costmap Fusion	51%	55%	58%	+7	Perception
Fleet Allocator	74%	79%	82%	+8	Fleet Services
Traffic Manager	65%	72%	74%	+9	Fleet Services
Charge Scheduler	54%	60%	63%	+9	Fleet Services
Safety Controller	82%	88%	91%	+9	Safety Engineering
E-Stop Chain	n/a	n/a	witnessed test	not comparable	Safety Engineering
Lidar Driver	37%	44%	44%	+7	Robot Firmware
Localization Engine	61%	66%	66%	+5	Autonomy Core
WMS Connector	60%	68%	69%	+9	Integrations
Order Broker	72%	75%	78%	+6	Fleet Services
Telemetry Ingest	55%	58%	61%	+6	Data and Telemetry
OTA Updater	67%	73%	76%	+9	Platform Infrastructure
Identity and Authorization	77%	82%	84%	+7	Security Engineering
Remote Operations Console	48%	51%	52%	+4	Applications
Simulation Harness	44%	46%	47%	+3	Developer Productivity
Hardware-in-the-Loop Rig	35%	36%	39%	+4	Robot Firmware
Fleet Health Monitor	50%	54%	57%	+7	Field Reliability
Battery Management Service	45%	51%	54%	+9	Robot Firmware
Gripper Controller	38%	38%	41%	+3	Robot Firmware
Map Authoring Studio	43%	48%	49%	+6	Applications

Service	Coverage 2025-09	Coverage 2026-03	Coverage at review	Change over period	Owning team
Site Provisioning Service	55%	62%	64%	+9	Platform Infrastructure
Event Bus	n/a	n/a	witnessed test	not comparable	Platform Infrastructure
Config Distribution	68%	69%	72%	+4	Platform Infrastructure
Robot Shadow Store	63%	68%	68%	+5	Fleet Services
Analytics Warehouse Loader	42%	43%	45%	+3	Data and Telemetry
Billing Usage Meter	75%	77%	79%	+4	Data and Telemetry
Alarm Router	46%	50%	51%	+5	Field Reliability
Video Snippet Service	37%	43%	43%	+6	Applications
Speed and Separation Monitor	82%	87%	87%	+5	Safety Engineering
Wheel Odometry Filter	53%	60%	62%	+9	Robot Firmware
Payload Scale Interface	29%	33%	36%	+7	Robot Firmware
Edge Gateway Agent	67%	73%	74%	+7	Platform Infrastructure
Release Orchestrator	47%	53%	55%	+8	Developer Productivity

6. What the incident record shows about the architecture

Harborline Systems felt Remote Operations Console incident INC-2025-078, which affected 4 site or tenant with no robot taken out of service. Following INC-2025-078 the team undertook to invalidate the roster cache on a fleet membership event and add a session-level regression test, with Anjali Sekhon accountable by 2025-08-31. Reported detection applies to stale state incident INC-2025-078, since an internal engineer noticed the discrepancy during a demonstration. Remote Operations Console incident INC-2025-078 was caused by a design gap: the session list cached the site roster and only invalidated it on a page reload. A contributing factor in Harborline Systems incident INC-2025-078 was that no test covered a roster change during an open operator session. stale state incident INC-2025-078 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

1 h 15 m elapsed before stale state event INC-2025-078 was mitigated, with resolution at 34 days. Incident INC-2025-078 was raised on 2025-06-24 after the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. Remote Operations Console incident INC-2025-078 classifies as a stale state event against Remote Operations Console. On delivery, Remote Operations Console incident INC-2025-078 shows this: landed on 2025-07-28; the regression test was still marked flaky at review. Stale state incident INC-2025-078 carries moderate on the cause and strong on the delivery position, which was checked in the repository evidence, and the reviewer declined to raise the finding beyond what that supports. stale state incident INC-2025-078 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The immediate event in algorithm defect event INC-2025-103 was that two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes, dated 2025-08-22. The reviewer's reading of incident INC-2025-103 is that the blast radius was set by a deployment topology decision made years earlier. Algorithm defect incident INC-2025-103 produced one commitment: to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, owned by Priya Raghunathan, dated 2025-10-31. Incident INC-2025-103 took 19 m to stabilise and 58 days to close out. Motion Planner incident INC-2025-103 belongs to the algorithm defect group and sits with Motion Planner. algorithm defect incident INC-2025-103 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

For INC-2025-103 the reviewer holds limited; the log retention window had closed before the reviewer asked for the raw data evidence and reports the finding at that strength. Ironwood Grocery Group incident INC-2025-103 is recorded as follows on delivery: the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. Ironwood Grocery Group incident INC-2025-103 was contained to one site and 2 robots. Ironwood Grocery Group incident INC-2025-103 would have been shorter or smaller had it not been that the simulation scenario for this geometry existed but had been excluded from the nightly suite. Underneath algorithm defect event INC-2025-103 lies the design

decision that the yield heuristic had no tie-break and both robots computed the same deference score. Motion Planner incident INC-2025-103 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The immediate event in modelling event INC-2026-057 was that state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating, dated 2026-05-19. Incident INC-2026-057 reached the engineering team when a robot derated mid-shift and the pack was pulled for analysis. The reviewer checked the Battery Management Service incident INC-2026-057 commitment against the code and release records; not landed; data collection is running and the imbalance alert is in review. Assessed on its own record, incident INC-2026-057 shows that the blast radius was set by a deployment topology decision made years earlier. Duration for incident INC-2026-057 splits into not a service outage; robots derated of impact and open at the review date to close the underlying cause. Battery Management Service incident INC-2026-057 was excluded from an earlier draft of TDR-2025-08 and reinstated after the reviewer obtained the supporting record.

Following Dunbar Industrial incident INC-2026-057 the team undertook to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, with Peter Halvorsen accountable by 2026-09-30. A contributing factor in incident INC-2026-057 was that the two affected sites moved to opportunity charging after the estimator had been validated. Classifying Dunbar Industrial incident INC-2026-057 as modelling matters because the Battery Management Service backlog contains two further items of the same kind. Modelling event INC-2026-057 rests on moderate on the cause and strong on the delivery position, which was checked in the repository evidence, which the reviewer states separately from the severity of the finding. So far as reach is concerned, Dunbar Industrial incident INC-2026-057 involved 2 sites and 5 robots. modelling event INC-2026-057 rests on a single source, and the reviewer has said so rather than implying corroboration.

Process gap incident INC-2025-096 is recorded as follows on delivery: the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. The two durations recorded for INC-2025-096 are not applicable; no service impact to mitigate and 26 days to resolve. The judgement recorded against incident INC-2025-096 is that the same class of defect can recur in two sibling components that were not changed. The classification applied to INC-2025-096 is process gap, and the affected component is Alarm Router. Internal incident INC-2025-096 began on 2025-08-05 when an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. process gap event INC-2025-096 was cross-checked against the engineering record supplied for TDR-2025-08, and the two agree.

The remediation committed after process gap incident INC-2025-096 was to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, owned by Wanda Njoku with a due date of 2025-10-15. Incident INC-2025-096 affected no production site and no robot directly, the exposure being to internal systems. Process gap event INC-2025-096 would have been shorter or smaller had it not been that the rotation source of truth was a spreadsheet maintained outside the alarm router. Reported detection applies to Alarm Router incident INC-2025-096, since discovered the next morning when the alarm was found unacknowledged. Alarm Router incident INC-2025-096 traces back to a single fact: the routing table fell back to a mailbox rather than to a secondary responder. process gap event INC-2025-096 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The immediate event in Kestrel Medical Supply incident INC-2026-004 was that relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site, dated 2026-01-13. INC-2026-004 traces back to a single fact: the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Incident INC-2026-004 closure status is not landed; the matcher replacement is in design and the re-survey is scheduled. Evidence strength on the performance incident INC-2026-004 analysis is moderate on the cause and strong on the delivery position, which was checked in the repository. INC-2026-004 reached the engineering team when the site's commissioning engineer raised it during acceptance testing. Localization Engine incident INC-2026-004 was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.

Performance is the class of Kestrel Medical Supply incident INC-2026-004; Localization Engine is the component. Contributing to Kestrel Medical Supply incident INC-2026-004, and not repaired by the remediation, is that the site was surveyed with a sparse feature set that increases cache misses. Drawing the evidence in Kestrel Medical Supply incident INC-2026-004 together, the blast radius was set by a deployment topology decision made years earlier. Kestrel Medical Supply felt performance event INC-2026-004, which affected one site and 18 robots. The remediation committed after INC-2026-004 was to replace the fallback matcher and re-survey the affected aisles, owned by Rahul Menon with a due

date of 2026-06-30. Kestrel Medical Supply incident INC-2026-004 is the clearest instance in TDR-2025-08 of a control that exists on paper and is weaker in practice.

The post-mortem for incident INC-2025-139 identifies the cause as follows: control and telemetry traffic shared one topic and one consumer group budget. Mitigation of INC-2025-139 took 2 h 47 m and full resolution took 63 days. On delivery, capacity event INC-2025-139 shows this: the split landed on 2026-01-04; the quarterly review has not yet run. After incident INC-2025-139, Erik Palmgren took the action to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly by 2026-01-31. Multiple sites incident INC-2025-139 carries a second-order finding: the topic inventory had recorded the sharing as acceptable in 2023 and it was never revisited. capacity incident INC-2025-139 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

2025-11-02 is the date of capacity event INC-2025-139, on which a telemetry backlog on a shared topic delayed control events by up to eleven seconds. Grading Event Bus incident INC-2025-139 on evidence rather than on consequence gives strong on the cause and limited on the contributing factor. Event Bus owns INC-2025-139, which the reviewer classes as capacity. The detection path for multiple sites incident INC-2025-139 was the traffic manager's own latency alert fired first. Multiple sites felt multiple sites incident INC-2025-139, which affected 7 sites and 246 robots. INC-2025-139 was raised again with the owning engineer after the initial interview, and the position did not change.

Underneath Crestline Fulfillment incident INC-2025-041 lies the design decision that the reservation lease had no expiry, so release depended on the departing robot. The trigger recorded for Crestline Fulfillment incident INC-2025-041 on 2025-03-14 was that a robot lost its network link mid-traverse and its corridor reservation was never released. The affected population in incident INC-2025-041 was 2 sites and 61 robots. Time to mitigate Crestline Fulfillment incident INC-2025-041 was 1 h 52 m, and time to resolve was 9 days. INC-2025-041 belongs to the resource leak group and sits with Traffic Manager. INC-2025-041 appears twice in the working papers for TDR-2025-08 because the first answer was corrected after a follow-up request.

Attention came to resource leak event INC-2025-041 when a site supervisor telephoned the on-call engineer after robots queued at one intersection. Evidence strength on the INC-2025-041 analysis is limited, being a single account corroborated only by the ticket timeline. Taken with the rest of the record, incident INC-2025-041 indicates that the failure was in the detection design rather than in the code that failed. For incident INC-2025-041 the follow-through was tested rather than assumed, and landed on 2025-05-22, verified in the reservation replay suite. Resource leak incident INC-2025-041 produced one commitment: to give every corridor reservation a lease with an expiry and add a site-level queue depth alert, owned by Owen Brantley, dated 2025-05-30. incident INC-2025-041 is the only entry in its group where the reviewer and the engineering team reached different conclusions.

7. Third-party components inside the reviewed services

The supply route for image-signing is public registry, signature verified. One point about image-signing is not resolved: attribution present. For image-signing the obligation is to attribute the component in the customer-facing notice file, and the reviewer tested whether it is met. image-signing sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Beyond licensing, scanmatch-legacy carries a single-maintainer dependency with no succession plan. Scanmatch-legacy is recorded as not approved in the component approval log. Scanmatch-legacy carries GPL-2.0 terms, classified strong for copyleft purposes. scanmatch-legacy is the entry that changed most between the first and second document requests behind TDR-2025-08.

Grpc-transport affects Event Bus; no other component reviewed depends on it. Version 1.62.0 of grpc-transport is the catalogued build at OSS-004, used service-to-service transport across the cloud. Compliance with the grpc-transport terms requires to share alike any derived asset it distributes. grpc-transport is one of the entries where TDR-2025-08 relies on the company's own measurement rather than an independent one.

Provenance of chartkit is public registry; upstream archived in 2025. Chartkit affects Remote Operations Console; no other component reviewed depends on it. Version 4.8.2 of chartkit is the catalogued build at OSS-006, used operator console visualisations. chartkit is the reason one section of TDR-2025-08 concludes narrowly rather than generally.

Sqlite-embedded presents a single-maintainer dependency with no succession plan to the engineering team. Linking mode matters for sqlite-embedded, which is used on-robot local state. For sqlite-embedded the obligation is nothing beyond retaining the licence text with the source, and the reviewer tested whether it is met. sqlite-embedded carries a follow-up request that was satisfied after the draft of TDR-2025-08 was circulated.

The approval position on jwt-verify is approved 2024-06-21. Version 5.2.1 of jwt-verify is the catalogued build at OSS-014, used token verification in the identity service. The terms applying to jwt-verify are MIT; the copyleft classification is no. jwt-verify would move from a concern to a verified defect if one further measurement were supplied.

OSS-026 covers csv-export at version 2.0.3, used billing and usage exports. Csv-export sits inside Billing Usage Meter and is therefore in the shipped product rather than in tooling. Csv-export carries the approval entry approved 2023-10-17; the reviewer looked for a corresponding decision record. csv-export was inspected in the repository rather than described from the ticket, which is why the account is specific.

Opencv-subset carries a practical obligation: to attribute the component in the customer-facing notice file. The terms applying to opencv-subset are Apache-2.0; the copyleft classification is no. Against opencv-subset the reviewer would add that notice reproduced in the robot documentation. opencv-subset is treated as material to the investment case rather than as a routine engineering item.

8. Architectural debt and the cost of removing it

The register entry programme item TD-22 reads: move the calibration record for the payload scale to the cloud, in the quality category. The business consequence recorded against TD-22 is that it calibration is lost on a compute module swap. About register item TD-22 it should be said that a small item that keeps returning as a field support cost. Payload Scale Interface item TD-22 was raised again with the owning engineer after the initial interview, and the position did not change.

Programme item TD-19 is unfunded and dependent on capital rather than engineering effort in the engineering plan the reviewer was shown. Programme item TD-19 is on the register because it the protective distance rests on a fixed approach speed assumption. TD-19 on Speed and Separation Monitor is in progress with a target date of 2026-08-31. safety item TD-19 was tested rather than accepted, and the test is described in the working papers behind TDR-2025-08.

Under safety, entry register item TD-20 is to validate and ship the wheel slip detector. TD-20 on Wheel Odometry Filter is not straightforward: the prototype uses a network-copyleft component that must be replaced before shipping. The business consequence recorded against safety item TD-20 is that it drift on wet floors is neither corrected nor flagged. TD-20 is one of the entries where TDR-2025-08 relies on the company's own measurement rather than an independent one.

Register item TD-06 carries an estimate of 1 engineer-quarter; the costed figure is $115,000. The reviewer's qualification on TD-06 on Simulation Harness is that blocked on the physics core determinism work. The dependency structure places register item TD-06 behind a supplier release outside the company's control. Simulation Harness item TD-06 is treated as material to the investment case rather than as a routine engineering item.

TD-18 on Safety Controller carries an estimate of 0.75 engineer-quarters; the costed figure is $104,000. Left in place, the condition behind programme item TD-18 investigation evidence is lost between weekly extractions. The reviewer's qualification on TD-18 on Safety Controller is that the safety team supports it; it is not funded in the current plan. register item TD-18 carries a follow-up request that was satisfied after the draft of TDR-2025-08 was circulated.

On funding, TD-08 is funded from the second quarter onward. The security supportability entry at programme item TD-08 is to replace the archived charting library in the operator console. The register shows Remote Operations Console item TD-08 as backlog, due 2027-01-31. security supportability item TD-08 would be a candidate for the first post-close engineering review under any reasonable plan.

9. Where the architecture meets the safety case

Hazard HZ-12 in the register produced two near-miss reports from site staff, and the register was not updated to reflect it. What the register does not say about Occasional-exposure hazard HZ-12, and the reviewer does, is that one site's dock layout leaves less clearance than the layout guide requires. The population exposed to Minor injury hazard HZ-12 is

every robot variant currently in production. hazard HZ-12 in the register is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

One caveat attaches to register entry HZ-06: one site had two lockout devices for eleven robots at the time of the visit. The design measure recorded at Occasional-exposure hazard HZ-06 is maintenance mode with a physical lockout and a two-step exit confirmation. On history, HZ-06 produced a single event before the current control was fitted. register entry HZ-06 carries a follow-up request that was satisfied after the draft of TDR-2025-08 was circulated.

Produced a single event before the current control was fitted is the incident history behind hazard HZ-07 in the register. The scope of Occasional-exposure hazard HZ-07 is robots fitted with the weighing deck. On the register's own scale, hazard HZ-07 is Equipment damage severity, Occasional exposure, Medium controllability. register entry HZ-07 carries a follow-up request that was satisfied after the draft of TDR-2025-08 was circulated.

Hazard HZ-08 in the register scores Equipment damage on severity; exposure is Frequent and controllability High. Equipment damage hazard HZ-08 applies to the manipulator-equipped variant only. Entry hazard HZ-08 in the hazard register reads: two robots collide at a corridor intersection. register entry HZ-08 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Lidar contamination degrades detection without a fault being raised is the hazard entered at register entry HZ-20. The parameter set for register entry HZ-20 reads Serious injury, Occasional, Medium. Residual risk at Occasional-exposure hazard HZ-20 is stated as Tolerable where cleaning compliance is evidenced, and the status is partially verified. Occasional-exposure hazard HZ-20 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Remote-exposure hazard HZ-04 is mitigated by force limiting with a mechanical release and an interlock on the access panel. Entry hazard HZ-04 in the register in the hazard register reads: gripper closes on a hand during a manual intervention. Register entry HZ-04 has never activated in the field and is evidenced only on the bench, which is the strongest available test of the control. register entry HZ-04 was cross-checked against the engineering record supplied for TDR-2025-08, and the two agree.

The reviewer's note on Serious injury hazard HZ-01 is that the fixed approach speed assumption is carried as an open qualification. Where Serious injury hazard HZ-01 is concerned, the hazard is that robot contacts a pedestrian while travelling in a shared aisle. Where HZ-01 bites is every site with a mezzanine or ramp. register entry HZ-01 illustrates why TDR-2025-08 separates the severity of a finding from the confidence behind it.

The three risk terms on hazard HZ-10 in the register are severity Fire, exposure Improbable and controllability Low. Evidence of verification at register entry HZ-10 consists of supplier test report plus an internal witnessed disconnect test. Hazard HZ-10 was scoped to the newest three sites, where commissioning is incomplete, and the reviewer checked the scoping against the fleet record. Fire hazard HZ-10 is not material on its own and belongs in this report because of what it reveals about the surrounding process.

10. Open items, declined remediations and qualifications

9 items in report TDR-2025-08 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. The open items in this report are grouped by what would resolve them rather than by which team owns them, because several cross team boundaries. Work on TDR-2025-08 ran over 6 weeks and included 17 interviews and read-only access to the engineering repositories. The working method behind the present report was to start from the incident record and work back to the design, rather than the other way round. The present report does not restate the company's own reporting where the reviewer was able to verify the underlying data instead.

One declinature in the present report rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. This report records 3 items the engineering team declined to remediate, each with the reason given and the reviewer's view of it. Selection for report TDR-2025-08 favoured components that appear in the incident record, which biases the sample towards known trouble and is stated as such. Report TDR-2025-08 sampled 16 services from an inventory of 34, weighted towards those on the critical control path. Report TDR-2025-08 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

S. Alderton notes in the present report that the company's own reporting is adequate for three of the four recommended indicators. 4 indicators are recommended in this report; the reviewer deliberately avoided proposing anything that needs a new pipeline. The instruction behind the present report was narrow: examine the service architecture, ownership boundaries and dependency structure of the fleet platform and report by 2025-08-19. Pinehurst Systems Assurance was instructed by Ardent Peak to examine the service architecture, ownership boundaries and dependency structure of the fleet platform; the resulting report is this report, dated 2025-08-19. The report at TDR-2025-08 is one input to a diligence process and is not a substitute for the company's own engineering judgement.

Where TDR-2025-08 says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Grading in the report at TDR-2025-08 is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. S. Alderton could not verify one class of claim in the report at TDR-2025-08, namely the behaviour of components deployed only at the two sites not visited. Where report TDR-2025-08 reports a supplier's test result, the reviewer read the certificate and did not witness the test. Pinehurst Systems Assurance has costed remediation at the company's own loaded engineering rate rather than at a market rate.

11. Conclusions and implications for the transaction

For Ardent Peak's purposes, the report at TDR-2025-08 concludes that the safety work is the strongest part of the engineering record and the delivery process is the weakest. Whatever weight a reader gives the individual findings, this report concludes that the safety work is the strongest part of the engineering record and the delivery process is the weakest. One limitation of the present report should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. The present report is limited by log retention: the telemetry window is ninety days, so events before 2026-01-05 could not be examined directly. Pinehurst Systems Assurance was given read access without restriction on scope within the agreed subject matter.

For the transaction, the present report identifies 2 matters the reviewer would raise before closing and 5 for the post-close engineering plan. S. Alderton would put 5 items from report TDR-2025-08 into the first hundred days rather than into a warranty. S. Alderton did not visit the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. The present report carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. S. Alderton interviewed 17 people for the present report; where an account was uncorroborated it is marked as such.

Monitoring recommended by the report at TDR-2025-08 relies on metrics the company already emits, so no new instrumentation is required to start. The monitoring pack the present report proposes would have surfaced two of the incidents examined here before a customer noticed. This report closes with 9 open items, listed with owner, evidence and the reviewer's view of whether the committed date is realistic. Where the report at TDR-2025-08 cannot conclude, it says what evidence would let it conclude. The report at TDR-2025-08 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.